№2: SIR -

2026-03-04

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- : Julia, DrWatson, Literate, Quarto

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- DrWatson

- DrWatson
- SIR - Julia

- DrWatson
- SIR - Julia
- SIR

- DrWatson
- SIR - Julia
- SIR
- Quarto Git

- Julia —

- Julia —
- DrWatson —

- Julia —
- DrWatson —
- DifferentialEquations.jl —

- Julia —
- DrWatson —
- DifferentialEquations.jl —
- Literate.jl —

- Julia —
- DrWatson —
- DifferentialEquations.jl —
- Literate.jl —
- Quarto — PDF/HTML

1:

:

```
cd ~/work/study/2026-1/2026-1--study--simulation-modeling/labs/lab02
```

1:

2: DrWatson

:

```
julia -e 'using Pkg; Pkg.add("DrWatson"); using DrWatson;
    initialize_project("project"; authors="                ", git=false)'
cd project
```

```
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02$ julia -e 'using Pkg; Pkg.add("DrWatson"); using DrWatson;
 initialize_project("project"; authors="Мария Улитина", git=false)'

    Updating registry at `~/.julia/registries/General.toml`
   Resolving package versions...
     Project No packages added to or removed from `~/.julia/environments/v1.12/Proj
ect.toml`
    Manifest No packages added to or removed from `~/.julia/environments/v1.12/Mani
fest.toml`
  Activating new project at `~/work/study/2026-1/2026-1-study-simulation-modeling/l
abs/lab02/project`
   Resolving package versions...
    Updating `~/work/study/2026-1/2026-1-study-simulation-modeling/labs/lab02/proje
ct/Project.toml`
```

2:

3:

add_packages.jl :

```
packages = ["DrWatson", "DifferentialEquations", "Plots", "DataFrames",
    "CSV", "JLD2", "Literate", "IJulia", "BenchmarkTools",
    "Quarto", "StatsPlots", "LaTeXStrings", "FFTW"]
Pkg.add(packages)
```

```
julia add_packages.jl
```

```
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02$ cd project
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano add_packages.jl
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ julia add_packages.jl
  Activating project at `~/work/study/2026-1/2026-1-study-simulation-modeling/labs/
lab02/project`
📦 Установка пакетов...
   Resolving package versions...
   Installed OffsetArrays ─────── v1.17.0
   Installed Distances ────────── v0.10.12
```

4:

:

```
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano scripts/tangle.jl
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano scripts/sir_ode.jl
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano scripts/sir_literate.jl
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano scripts/lv_literate.jl
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano scripts/sir_parametric.jl
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano preamble.tex
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano report.qmd
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ nano Makefile
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ make all
julia --project=. scripts/sir_ode.jl
✅ SIR базовый выполнен
julia --project=. scripts/tangle.jl scripts/sir_literate.jl
```

4:

5: git

, git

```
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ git add .
mmulitina@mmulitina-VMware-Virtual-Platform:~/work/study/2026-1/2026-1-study-simula
tion-modeling/labs/lab02/project$ git commit -m "Выполнена ЛР №2: SIR и Лотки-Вольт
ерры"
```

5: git

- make all

- `make all`
- SIR - Julia

- `make all`
- SIR - Julia
- β

- `make all`
- SIR - Julia
- β
- Quarto PDF

- `make all`
- SIR - Julia
- β
- Quarto PDF
- Git-